

Project Status Report

Socio-Economic & Demographic Analysis of Indian Festivals · DAV Laboratory mini project · B.E. Sem VII
ECS, Vidyalankar Institute of Technology
Team: Jainam Jain (23108B0084), Vrushan Patil, Dhanush Chowke, Aditya Tambe, Vivek Jaiswal · Guide: Prof.
Uma Jaishankar · Status as of 05 October 2026

1. Status at a glance

Task	Status	What exists
1. Data collection & preprocessing	Done	2,705 festival-holiday records, 99 festivals, 29 states/UTs, 2024–26; 78 verified spending and crowd figures; Census 2011 religion data; data dictionary
2. Machine learning	Done	3 models (clustering, classification, regression) with code, saved model files and evaluation metrics
3. Statistics & EDA	Done	15 hypothesis tests, 10 validation tables, 10 statistical charts, summary report
4a. Blog article	Done	Published on Medium on 28 Sep 2026; 9 charts, 9 cited sources, 4 interactive charts
4b. External engagement	In progress	0 of 40 external reactions/comments logged; 0 comments answered
5. Video & repository	Done	Narrated 1080p video (about 5.4 min); public GitHub repository; 8-page technical report

2. What was done in each task

Task 1: Data

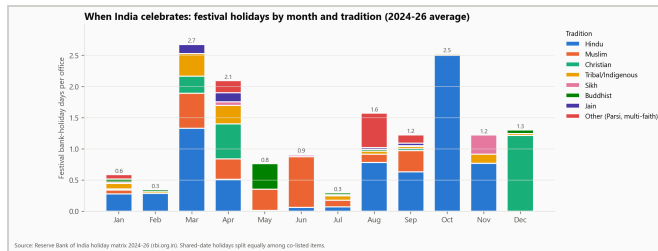
- Scraped the Reserve Bank of India's official bank-holiday lists for all 34 regional offices, every month of 2024, 2025 and 2026 (2,031 office-date rows, source URL on every row).
- Resolved RBI's holiday labels into 99 named festivals across 9 tradition groups: Hindu, Muslim, Christian, Sikh, Buddhist, Jain, Parsi, tribal/indigenous and multi-faith cultural.
- Took each festival's tradition from its Wikipedia infobox, or a Government portal where no article exists.
- Added Census of India 2011 religion-by-state data and 78 spending and crowd figures, each opened and confirmed on its source page. No synthetic records.
- Known limit: RBI gives one combined label per date, so shared dates are split equally; 39 festivals are flagged as "reach not identifiable".

Task 2: Machine learning

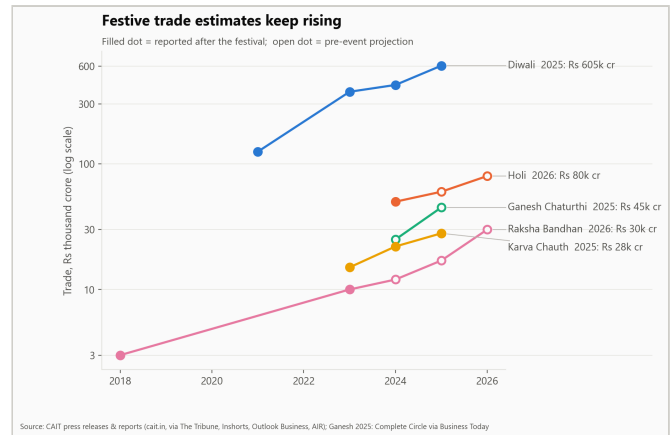
- State clustering (K-Means):** 4 groups, silhouette 0.41, bootstrap stability 0.61. Groups do not follow geography (agreement with regions 0.03).
- Tradition classifier (Random Forest):** macro-F1 0.40 against 0.16 for a majority-class baseline, repeated 5-fold cross-validation.
- Festive-trade regression:** growth 36% a year in this model (34% in the Task 3 test, which also uses the 2026 figures), leave-one-out error 20%; predicted the 2026 Holi figure within 3% and Raksha Bandhan within 23%.

Task 3: Statistics

- Festival holidays are concentrated by month (chi-square = 551, $p < 0.001$); peaks in March, October and April.
- Each faith's share of a state's holidays rises with its population share (Spearman 0.42 to 0.71, all significant).
- Hindu festivals are about 46% of festival holidays against 79.8% of the population, because several minority festivals are recognised almost nationwide.
- No significant difference between the six regions ($p = 0.58$); a festival's season is independent of its tradition ($p = 0.58$).



Festival holidays by month and tradition



Festive trade estimates by year

Task 4: Blog

- Article: "What India's Bank-Holiday Calendar Reveals About How the Country Celebrates", credited to all five team members, neutral and secular framing, every figure cited.
- Live at: <https://medium.com/@ijainamjain/what-indias-bank-holiday-calendar-reveals-about-how-the-country-celebrates-97c99917b2d3>
- Engagement audit tooling is ready: a log (one row per external reaction or comment, with screenshot) and a script that counts only people outside VIT.

Task 5: Video and repository

- Video: 11 narrated slides covering motivation, data integrity, statistics, ML, economics, blog reception and takeaways. The reception slide updates from the engagement log.
- Repository (code, datasets, models, charts, report, video): <https://github.com/ijainamjain/indian-festivals-data-analysis>

3. What is still pending

External engagement: 0 of 40 logged. This is the only open requirement.

- All five members share the article outside VIT (LinkedIn, professional and interest groups) and reply to every comment.
- Record each external reaction or comment in `blog/engagement/engagement_log.csv` with a screenshot.
- Re-run scripts 09, 10 and 11 so the audit report, video and technical report show the final counts.
- Optional: add the other four members' roll numbers to the technical report.

4. Where everything is

Item	Location (inside the project folder)
Raw and cleaned datasets, data dictionary	data/raw, data/processed, data/data_dictionary.md
Code (12 pipeline scripts, 2 helper modules)	code/01_... to code/12_..., festival_catalog.py, viz_style.py
Statistics tables and charts	outputs/stats, outputs/figures
ML metrics and saved models	outputs/ml, outputs/models
Blog source, images, interactive charts	blog/blog_post.md, blog/images, blog/interactive
Engagement log and audit	blog/engagement
Technical report (8 pages)	report/DAV Project Report_23108B0084.pdf
Video	video/Indian_Festivals_Project_Report.mp4